OBJECTIVE TESTS – 26

1. The basic requirement of a d.c. armature winding is that it must be
   (*a*) a closed one
   (*b*) a lap winding
   (*c*) a wave winding
   (*d*) either (*b*) or (*c*)
2. A wave winding must go at least .... around the armature before it closes back where it started.
   (*a*) once
   (*b*) twice
   (*c*) thrice
   (*d*) four times
3. The d.c. armature winding in which coil sides are a pole pitch apart is called .... winding.
   (*a*) multiplex
   (*b*) fractional-pitch
   (*c*) full-pitch
   (*d*) pole-pitch
4. For making coil span equal to a pole pitch in the armature winding of a d.c. generator, the back pitch of the winding must equal the number of
   (*a*) commutator bars per pole
   (*b*) winding elements
   (*c*) armature conductors per path
   (*d*) armature parallel paths.
5. The primary reason for making the coil span of a d.c. armature winding equal to a pole pitch is to
   (*a*) obtain a coil span of 180° (electrical)
   (*b*) ensure the addition of e.m.fs. of consecutive turns
   (*c*) distribute the winding uniformly under different poles
   (*d*) obtain a full-pitch winding.
6. In a 4-pole, 35 slot d.c. armature, 180 electrical-degree coil span will be obtained when coils occupy .... slots.
   (*a*) 1 and 10
   (*b*) 1 and 9
   (*c*) 2 and 11
   (*d*) 3 and 12
7. The armature of a d.c. generator has a 2-layer lap-winding housed in 72 slots with six conductors/slot. What is the minimum number of commutator bars required for the armature?
   (*a*) 72
   (*b*) 432
   (*c*) 216
   (*d*) 36
8. The sole purpose of a commutator in a d.c. Generator is to
   (*a*) increase output voltage
   (*b*) reduce sparking at brushes
   (*c*) provide smoother output
   (*d*) convert the induced a.c. into d.c.
9. For a 4-pole, 2-layer, d.c., lap-winding with 20 slots and one conductor per layer, the number of commutator bars is
   (*a*) 80
   (*b*) 20
   (*c*) 40
   (*d*) 160
10. A 4-pole, 12-slot lap-wound d.c. armature has two coil-sides/slot. Assuming single turn coils and progressive winding, the back pitch would be
    (*a*) 5
    (*b*) 7
    (*c*) 3
    (*d*) 6
11. If in the case of a certain d.c. armature, the number of commutator segments is found either one less or more than the number of slots, the armature must be having a simplex .... winding.
    (*a*) wave
    (*b*) lap
    (*c*) frog leg
    (*d*) multielement
12. Lap winding is suitable for .... current, ... voltage d.c. generators.
    (*a*) high, low
    (*b*) low, high
    (*c*) low, low
    (*d*) high, high

13. The series field of a short-shunt d.c. generator is excited by .... currents.
(a) shunt
(b) armature
(c) load
(d) external

14. In a d.c. generator, the generated e.m.f. is directly proportional to the
(a) field current
(b) pole flux
(c) number of armature parallel paths
(d) number of dummy coils

15. In a 12-pole triplex lap-wound d.c. armature, each conductor can carry a current of 100 A. The rated current of this armature is .... ampere.
(a) 600
(b) 1200
(c) 2400
(d) 3600

16. The commercial efficiency of a shunt generator is maximum when its variable loss equals .... loss.
(a) constant
(b) stray
(c) iron
(d) friction and windage

17. In small d.c. machines, armature slots are sometimes not made axial but are skewed. Though skewing makes winding a little more difficult, yet it results in
(a) quieter operation
(b) slight decrease in losses
(c) saving of copper
(d) both (a) and (b)

18. The critical resistance of the d.c. generator is the resistance of
(a) armature
(b) field
(c) load
(d) brushes (Grad. I.E.T.E Dec. 1985)

ANSWERS

1. (a) 2. (b) 3. (c) 4. (a) 5. (b) 6. (b) 7. (c) 8. (d) 9. (b) 10. (b)
11. (a) 12. (a) 13. (c) 14. (b) 15. (d) 16. (a) 17. (d) 18. (b)

OBJECTIVE TEST – 27

1. In d.c. generators, armature reaction is produced actually by
   (*a*) its field current
   (*b*) armature conductors
   (*c*) field pole winding
   (*d*) load current in armature
2. In a d.c. generator, the effect of armature reaction on the main pole flux is to
   (*a*) reduce it (*b*) distort it
   (*c*) reverse it (*d*) both (*a*) and (*b*)
3. In a clockwise-rotating loaded d.c. generator, brushes have to be shifted
   (*a*) clockwise
   (*b*) counterclockwise
   (*c*) either (*a*) or (*b*)
   (*d*) neither (*a*) nor (*b*).
4. The primary reason for providing compensating windings in a d.c. generator is to
   (*a*) compensate for decrease in main flux
   (*b*) neutralize armature mmf
   (*c*) neutralize cross-magnetising flux
   (*d*) maintain uniform flux distribution.
5. The main function of interpoles is to minimize .... between the brushes and the commutator when the d.c. machine is loaded.
   (*a*) friction (*b*) sparking
   (*c*) current (*d*) wear and tear

6. In a 6-pole d.c. machine, 90 mechanical degrees correspond to .... electrical degrees.
   (a) 30 (b) 180
   (c) 45 (d) 270
7. The most likely cause(s) of sparking at the brushes in a d.c. machine is /are
   (a) open coil in the armature
   (b) defective interpoles
   (c) incorrect brush spring pressure
   (d) all of the above
8. In a 10-pole, lap-wound d.c. generator, the number of active armature conductors per pole is 50. The number of compensating conductors per pole required is
   (a) 5 (b) 50
   (c) 500 (d) 10
9. The commutation process in a d.c. generator basically involves
   (a) passage of current from moving armature to a stationary load
   (b) reversal of current in an armature coil as it crosses MNA
   (c) conversion of a.c. to d.c.
   (d) suppression of reactance voltage
10. Point out the WRONG statement. In d.c. generators, commutation can be improved by
    (a) using interpoles
    (b) using carbon brushes in place of Cu brushes
    (c) shifting brush axis in the direction of armature rotation
    (d) none of the above
11. Each of the following statements regarding interpoles is true except
    (a) they are small yoke-fixed poles spaced in between the main poles
    (b) they are connected in parallel with the armature so that they carry part of the armature current
    (c) their polarity, in the case of generators is the same as that of the main pole ahead
    (d) they automatically neutralize not only reactance voltage but cross-magnetisation as well
12. Shunt generators are most suited for stable parallel operation because of their voltage characteristics.
    (a) identical (b) dropping
    (c) linear (d) rising
13. Two parallel shunt generators will divide the total load equally in proportion to their kilowatt output ratings only when they have the same
    (a) rated voltage
    (b) voltage regulation
    (c) internal I_aR_a drops
    (d) boths (a) and (b)
14. The main function of an equalizer bar is to make the parallel operátion of two over-compounded d.c. generators
    (a) stable (b) possible
    (c) regular (d) smooth
15. The *essential* condition for stable parallel operation *A* Two d.c. generators having similar characteristics is that they should have
    (a) same kilowatt ouput ratings
    (b) droping voltage characterisitcs
    (c) same percentage regulation
    (d) same no-load and full-load speed
16. The main factor which loads to unstable parallel operation of flat-and over-compound d.c. generators is
    (a) unequal number of turns in their series field windings
    (b) unequal series field resistances
    (c) their rising voltage characteristics
    (d) unequal speed regulation of their prime movers
17. The simplest way to shift load from one d.c. shunt generator running in parallel with another is to
    (a) adjust their field rheostats
    (b) insert resistance in their armature circuits
    (c) adjust speeds of their prime movers
    (d) use equalizer connections
18. Which one of the following types of generators does NOT need equalizers for satisfactory parallel operation ?
    (a) series (b) over-compound
    (c) flat-compound (d) under-compound.

ANSWERS

1. (d) 2. (d) 3. (a) 4. (c) 5. (b) 6. (d) 7. (d) 8. (a) 9. (b) 10. (d) 11. (b) 12. (b) 13. (d) 14. (a) 15. (b) 16. (c) 17. (a) 18. (d)

OBJECTIVE TESTS – 28

1. The external characteristic of a shunt generator can be obtained directly from its —— characteristic.
   (*a*) internal (*b*) open-circuit
   (*c*) load-saturation (*d*) performance
2. Load saturation characteristic of a d.c. generator gives relation between
   (*a*) V and I_a (*b*) E and I_a
   (*c*) E_0 and I_f (*d*) V and I_f
3. The slight curvature at the lower end of the *O.C.C.* of a self-excited d.c. generator is due to
   (*a*) residual pole flux
   (*b*) high armature speed
   (*c*) magnetic inertia
   (*d*) high field circuit resistance.
4. For the voltage built-up of a self-excited d.c. generator, which of the following is not an essential condition ?
   (*a*) There must be some residual flux
   (*b*) Field winding mmf must aid the residual flux
   (*c*) Total field circuit resistance must be less than the critical value
   (*d*) Armature speed must be very high.
5. The voltage build-up process of a d.c. generator is
   (*a*) difficult (*b*) delayed
   (*c*) cumulative (*d*) infinite
6. Which of the following d.c. generator cannot build up on open-circuit ?
   (*a*) shunt (*b*) series
   (*c*) short shunt (*d*) long shunt
7. If a self-excited d.c. generator after being installed, fails to build up on its first trial run, the first thing to do is to
   (*a*) increase the field resistance
   (*b*) check armature insulation
   (*c*) reverse field connections
   (*d*) increase the speed of prime mover.
8. If residual magnetism of a shunt generator is destroyed accidentally, it may be restored by connecting its shunt field
   (*a*) to earth
   (*b*) to an a.c. source
   (*b*) in reverse
   (*d*) to a d.c. source.
9. The three factors which cause decrease in the terminal voltage of a shunt generator are
   (*a*) armature reactances
   (*b*) armature resistance
   (*c*) armature leakages
   (*d*) armature reaction
   (*e*) reduction in field current
10. If field resistance of a d.c. shunt generator is increased beyond its critical value, the generator
    (*a*) output voltage will exceed its name-plate rating
    (*b*) will not build up

(c) may burn out if loaded to its name-plate rating

(d) power output may exceed its name-plate rating

11. An ideal d.c. generator is one that has .... voltage regulation.

(a) low (b) zero

(c) positive (d) negative

12. The .... generator has poorest voltage regulation.

(a) series (b) shunt

(c) compound (d) high

13. The voltage regulation of an overcompound d.c. generator is always ....

(a) positive

(b) negative

(c) zero

(d) high

14. Most commercial compound d.c. generator are normally supplied by the manufacturers as over compound machines because

(a) they are ideally suited for transmission of d.c. energy to remotely-located loads

(b) degree of compounding can be adjusted by using a divertor across series field

(c) they are more cost effective than shunt generators

(d) they have zero percent regulation.

ANSWERS

1. (b) 2. (d) 3. (c) 4. (d) 5. (c) 6. (b) 7. (c) 8. (d) 9. (b,d,e) 10. (b) 11. (b) 12. (a) 13. (b) 14. (b).